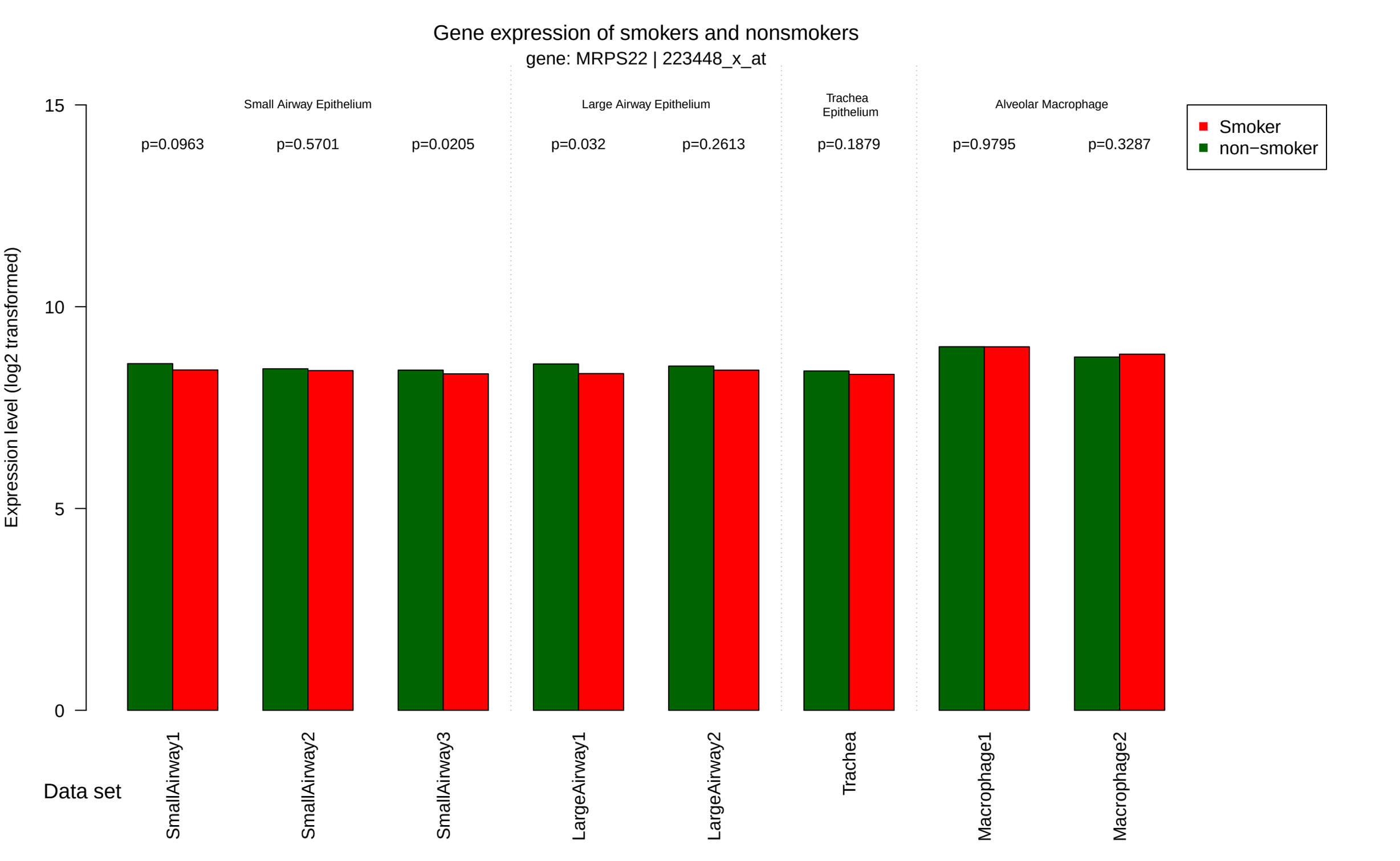
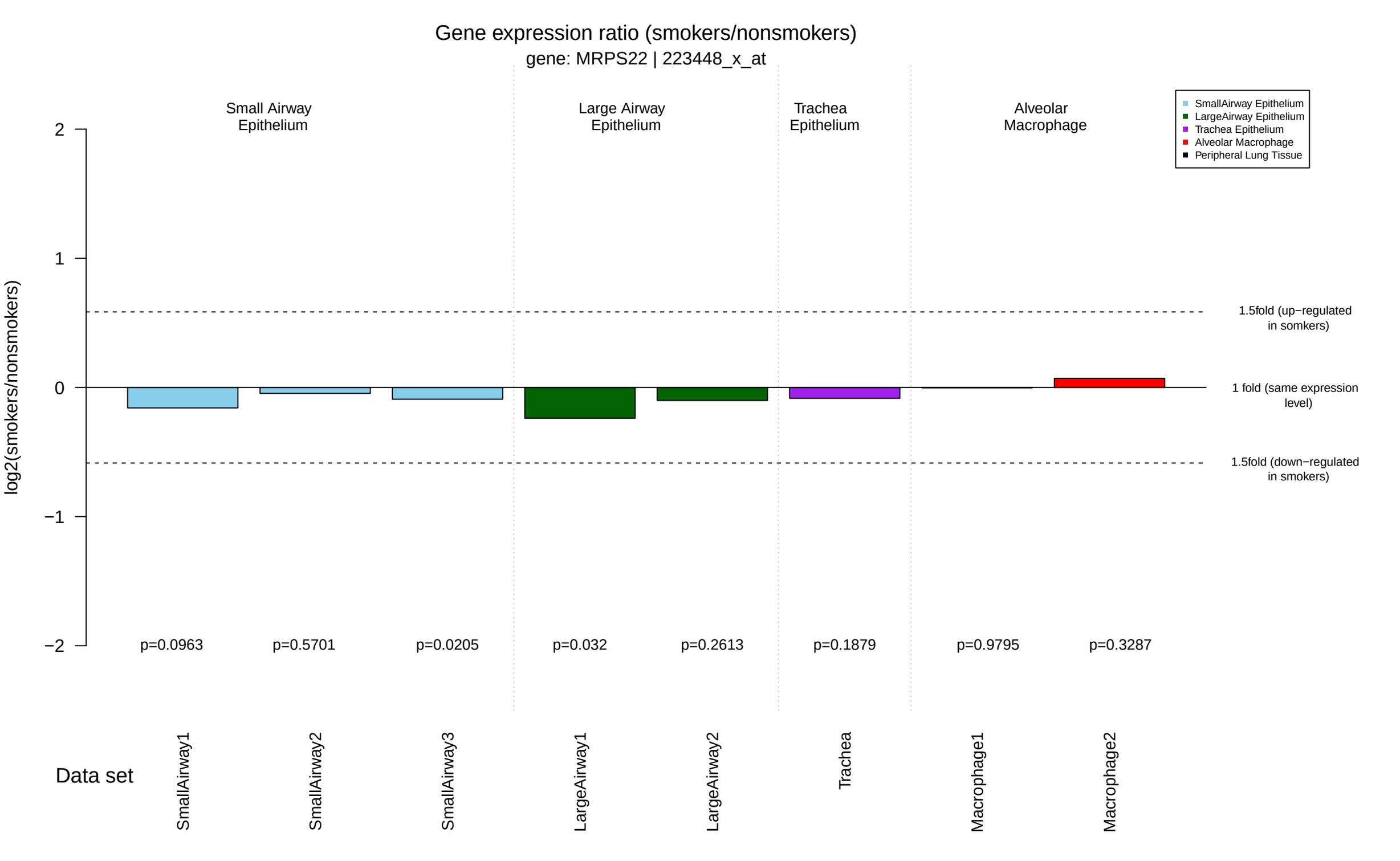
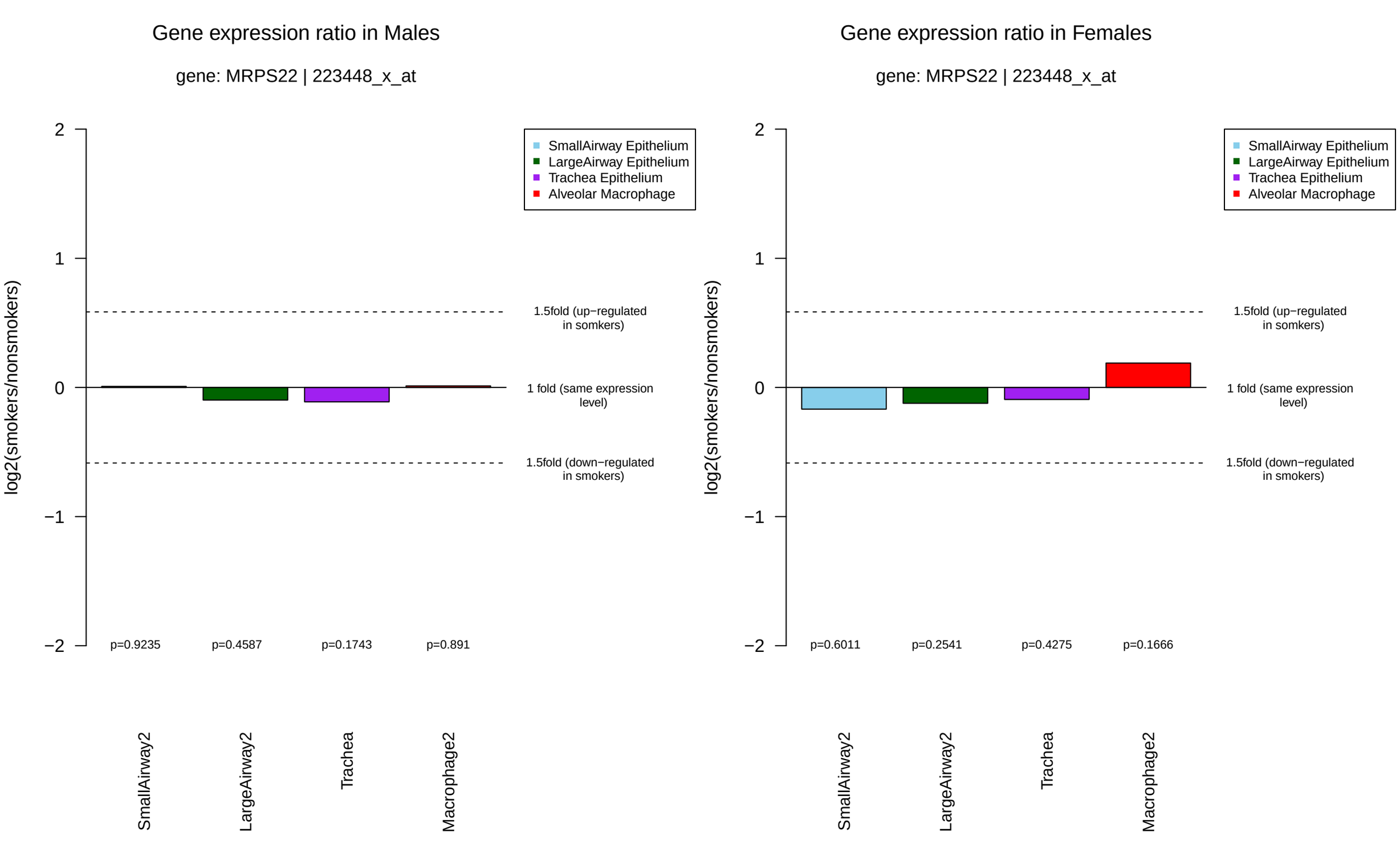
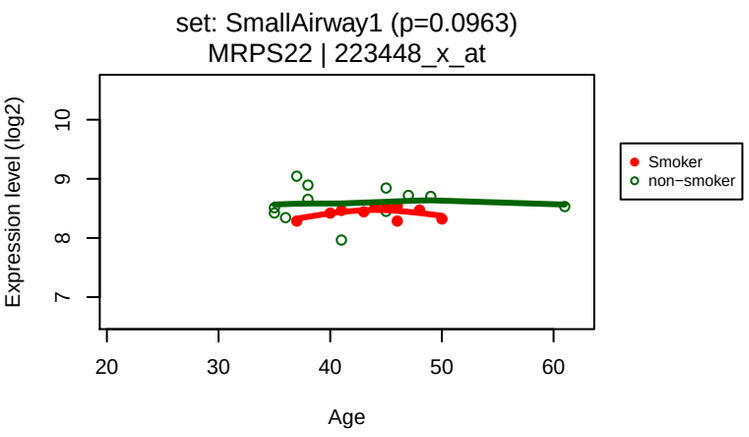
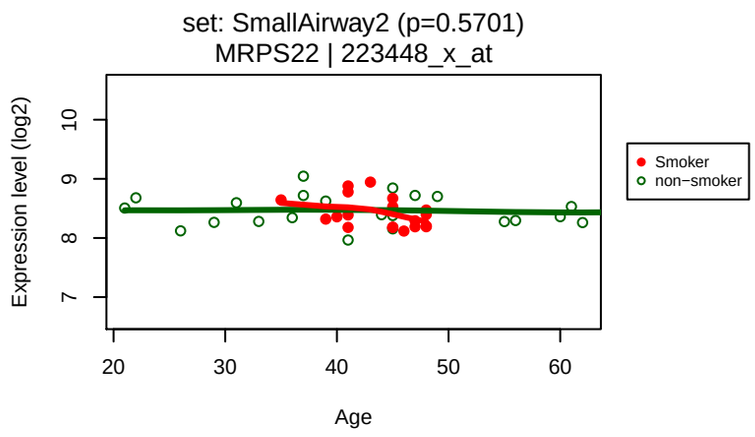
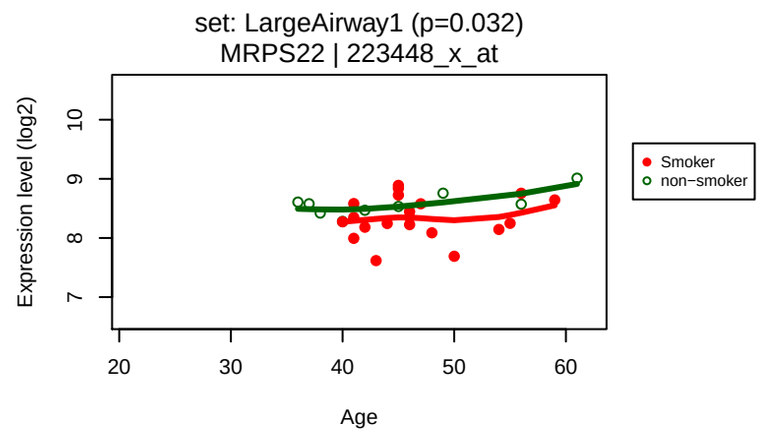

Gene expression of smokers and nonsmokers
gene: MRPS22 | 223448_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0963
p=0.5701
p=0.0205
p=0.032
p=0.2613
p=0.1879
p=0.9795
p=0.3287
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

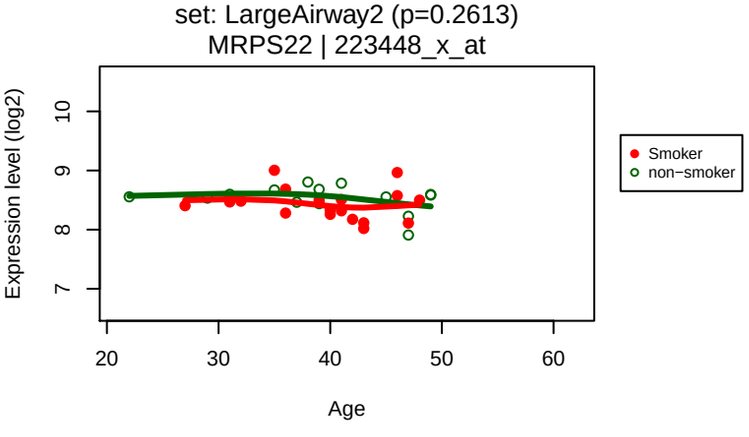

Gene expression ratio (smokers/nonsmokers)
gene: MRPS22 | 223448_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0963
p=0.5701
p=0.0205
p=0.032
p=0.2613
p=0.1879
p=0.9795
p=0.3287
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

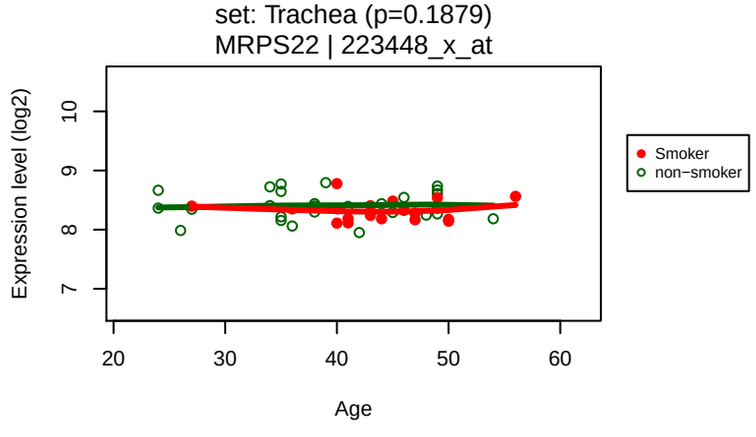

Gene expression ratio in Males
gene: MRPS22 | 223448_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9235
p=0.4587
p=0.1743
p=0.891
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPS22 | 223448_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6011
p=0.2541
p=0.4275
p=0.1666
SmallAirway2
LargeAirway2
Trachea
Macrophage2

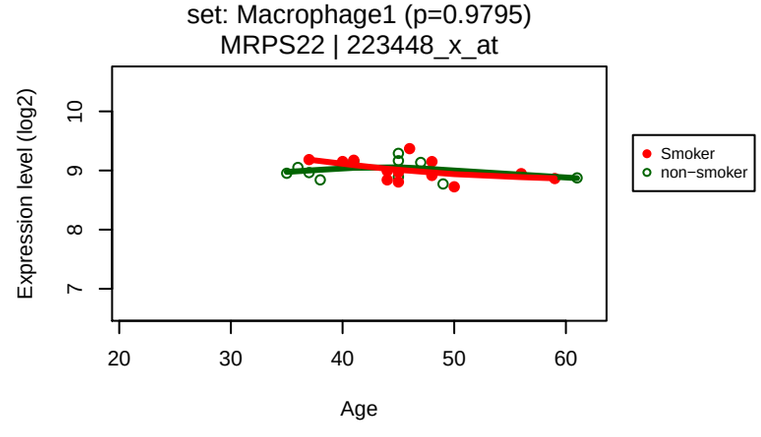

set: SmallAirway1 (p=0.0963)
MRPS22 | 223448_x_at
Expression level (log2)
Age
Smoker
non−smoker

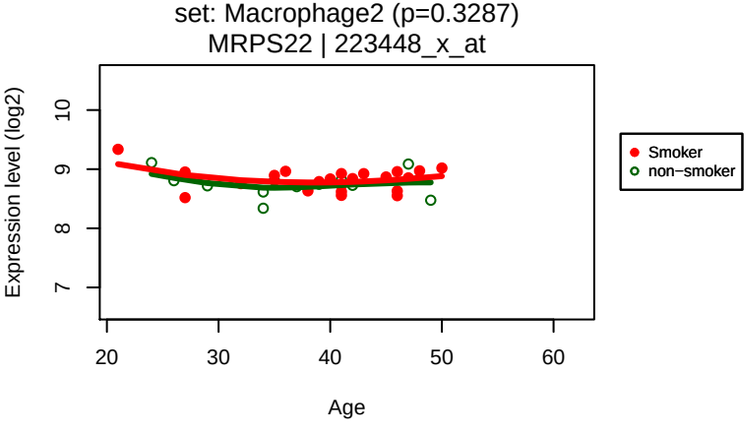

set: SmallAirway2 (p=0.5701)
MRPS22 | 223448_x_at
Expression level (log2)
Age
Smoker
non−smoker

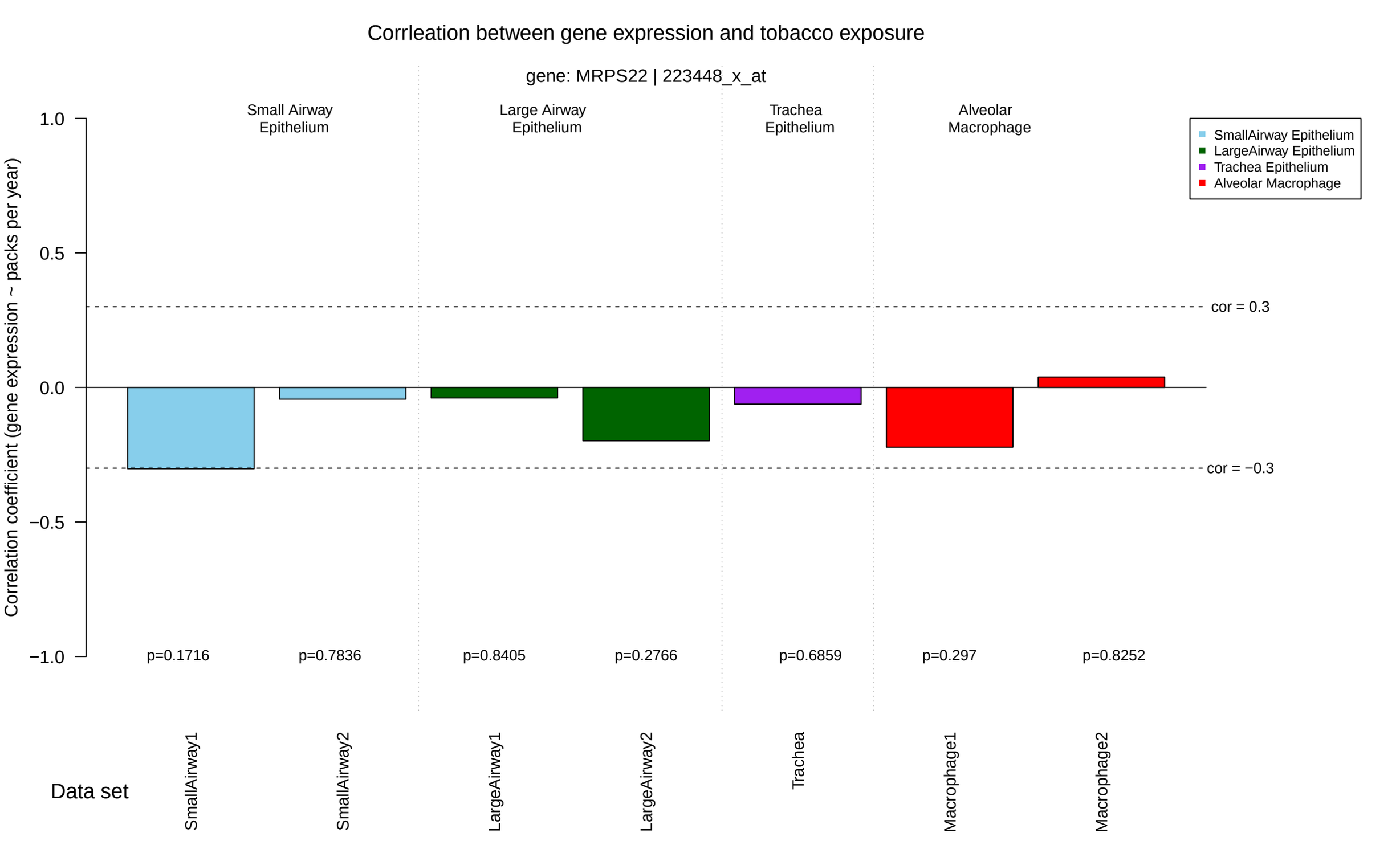

set: LargeAirway1 (p=0.032)
MRPS22 | 223448_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2613)
MRPS22 | 223448_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1879)
MRPS22 | 223448_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9795)
MRPS22 | 223448_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3287)
MRPS22 | 223448_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPS22 | 223448_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1716
p=0.7836
p=0.8405
p=0.2766
p=0.6859
p=0.297
p=0.8252
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2